

Assignment 2 - DS3214

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Problem: Obtaining a diabetes threshold from glucose data

We aim to determine an optimal threshold value that can classify patients as diabetic or non-diabetic. We will employ three clustering algorithms- K-means clustering, DBSCAN and Gaussian Mixture Models (GMM). We will then examine various sample sizes ($n = 20, 50$ and 100) to check whether that makes any difference?

Dataset

We will use some simulated data for our experiment. A dataset is created containing glucose values that follow two distinct normal distributions:

1. Normal glucose range: Gaussian distribution with a mean of 80 mg/dL and standard deviation of 20 mg/dL,
2. Hyperglycaemic range: Gaussian distribution with a mean of 220 mg/dL and standard deviation of 50 mg/dL.

Labelled sample data of size n is generated from each of the normal distributions mentioned above. The data obtained from both the distributions is then concatenated and shuffled to create the final dataset of size twice n . Here $n = 20, 50$ and 100 . Refer Figure1 for data distribution plots.

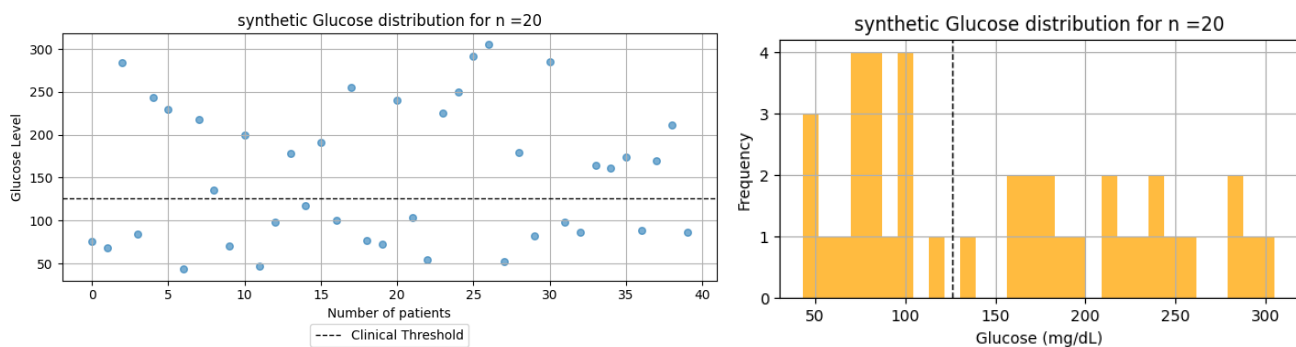


Figure1 (a) Synthetic glucose distribution (original dataset) for sample size 20 each

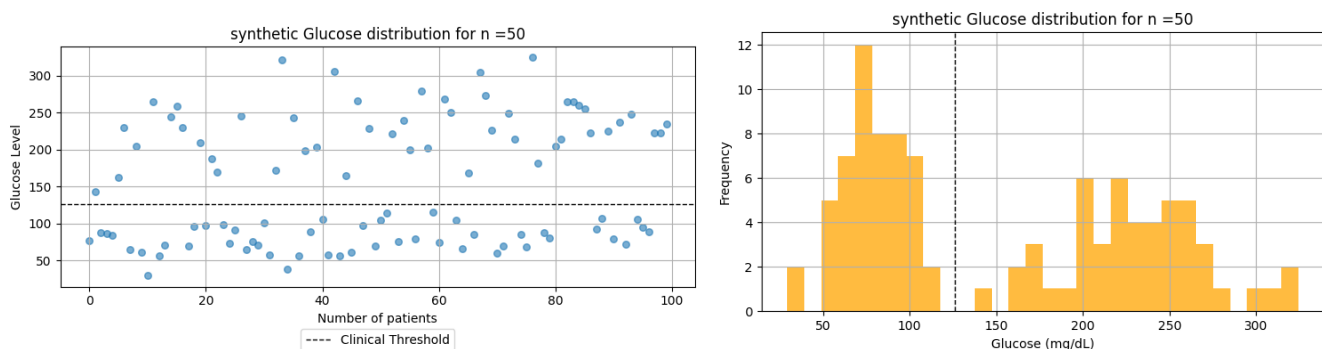


Figure1 (b) Synthetic glucose distribution (original dataset) for sample size 50 each

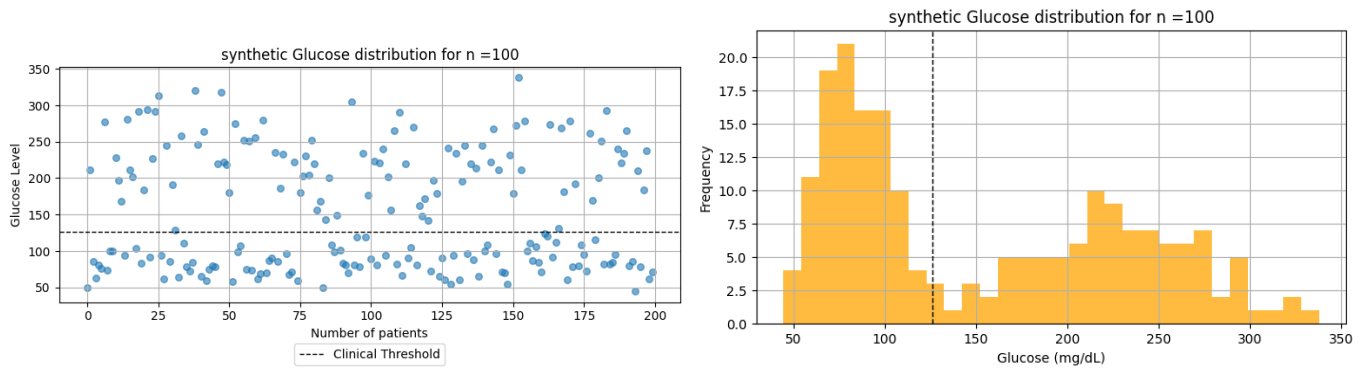


Figure1 (c) Synthetic glucose distribution (original dataset) for sample size 100 each

Clustering results and analyses:

1. K-means Clustering-

It is the simplest approach and requires specifying the number of clusters formed before-hand. For this dataset the number of clusters is set to 2 and random state iterations is set to 100. The threshold obtained from this approach varies between 150 and 160 mg/dL.

K-means is able to cluster but the threshold is high as compared to the clinical threshold of 126 mg/dL. K-means finds two centroids (means) based on Euclidian distance and finds a mathematical centre not a biological boundary. It also ignores the borderline patients. **Therefore, K-means is not an ideal clustering algorithm in this situation.**

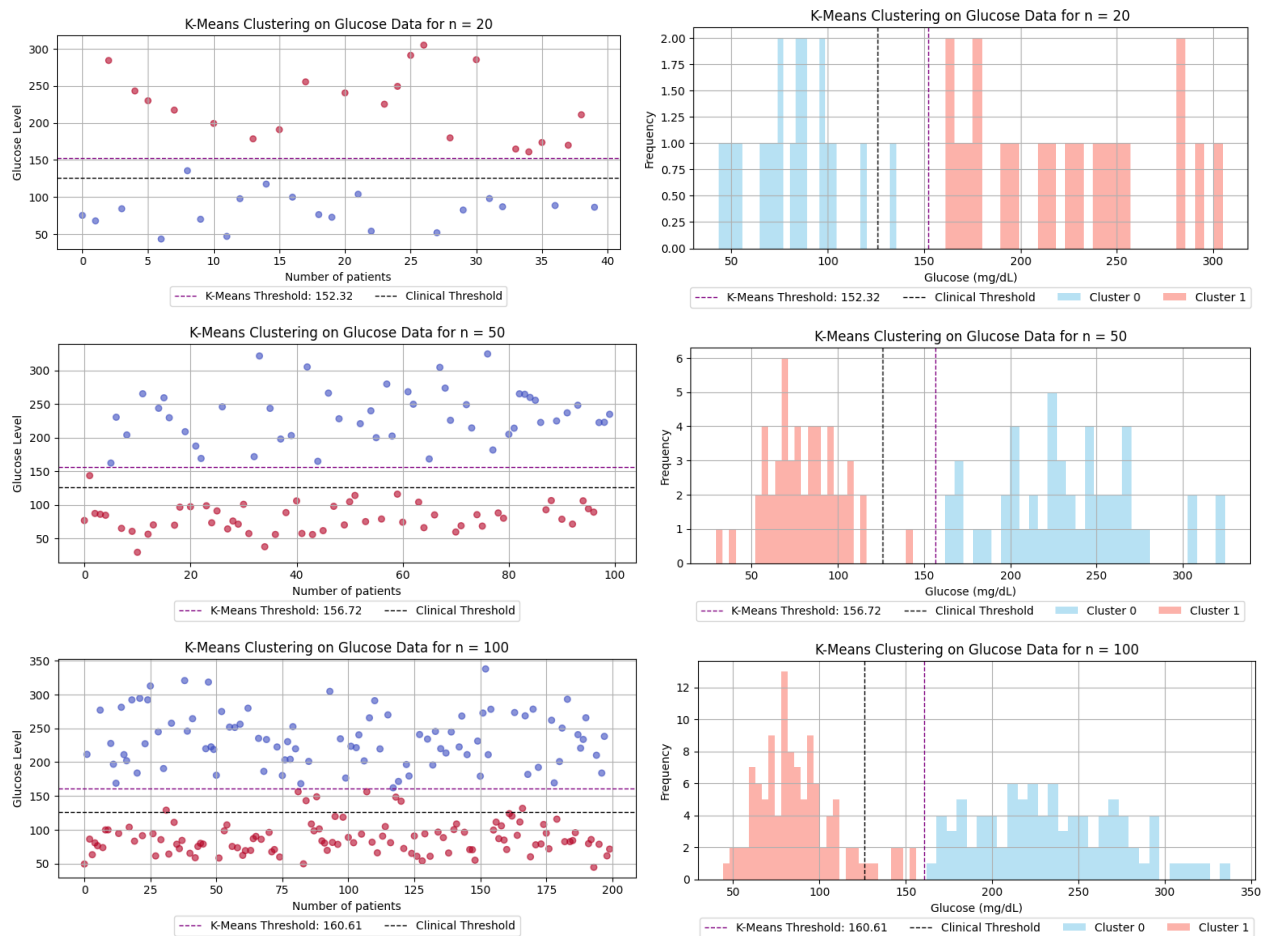


Figure 2 : Distributions and scatter plots displaying the identified clusters and the calculated thresholds after applying K-means clustering for n = 20 (**152.32 mg/dL**), 50 (**156.72 mg/dL**) and 100 (**160.61 mg/dL**). Numbers in bracket are the K-means thresholds for respective n.

2. DBSCAN Clustering-

It is a density based clustering algorithm that groups together points that are closely packed (i.e., in regions of high density) and marks points in low-density regions as outliers/noise. It doesn't require the user to specify the number of clusters before-hand rather it forms clusters by itself based on the epsilon (Maximum distance between two samples to be considered "neighbors") and minimum samples (Minimum number of neighbors required to form a core point). In our case we have chosen fixed $\epsilon = 0.12$ and $\text{min_samples} = 6$.

DBSCAN **fails** to give valid clusters and a valid threshold as on every iteration (with different seed and different parameters) it gives different number of clusters. Since, DBSCAN looks for **low-density "gaps"** to separate clusters, in 1D data, this density dip is often **too shallow**, so DBSCAN either forms one big cluster + noise (if ϵ is too small) or divides the data into many clusters (if ϵ is too large). The parameters depend heavily on the sample sizes in our case hence, to get 2 valid clusters we might have to do a grid search to find the best combination of parameters or use different set of parameters for different sample sizes. **Due its inability to give biologically valid clusters (in this case 2 clusters are required one for normal and other for diabetic) DBSCAN is a very poor clustering algorithm for this situation.**

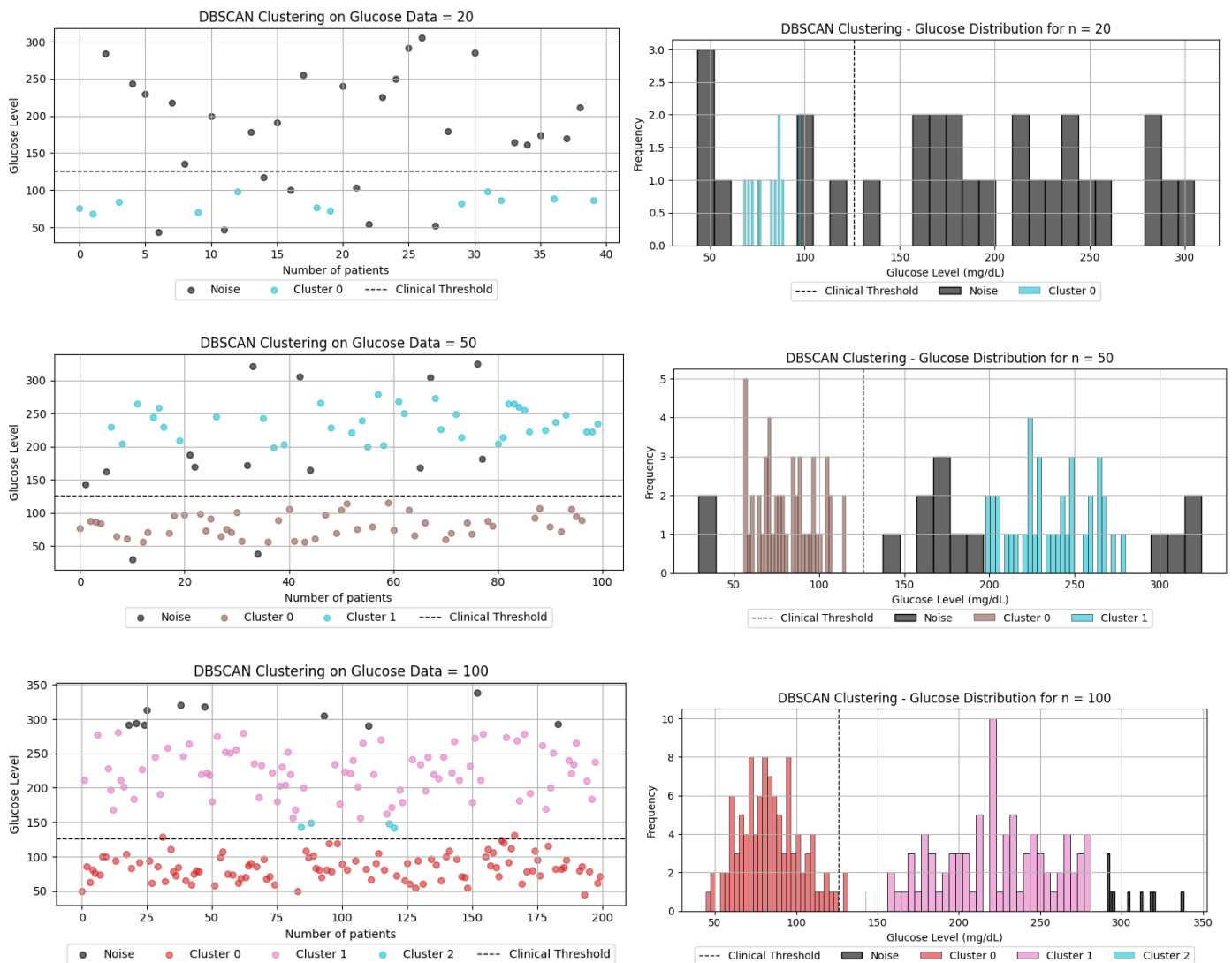


Figure 3: Distributions and scatter plots displaying the identified clusters after applying DBSCAN clustering for n = 20, 50 and 100

3. GMM-EM Clustering:

It is a probabilistic model that assumes data is generated from a mixture of multiple gaussian distributions. It uses Expectation-Maximization (EM) algorithm to fit GMM to the data. In our case the data is mixture of two gaussians and the initial starting weights are set to 0.5 for each. GMM gives a threshold between 120 and 130 gm/dL.

Since the data we use is a gaussian mixture which is what GMM assumes the data to be, it captures both the tight, low-variance Gaussian for healthy individuals as well as the broad, high-variance Gaussian for diabetics and hence is it not only able to give valid clusters but also gives a threshold very close to the clinical threshold of 126 mg/dL. **Therefore, GMM is the best clustering method so far when it comes to Gaussian-like biological data like ours.**

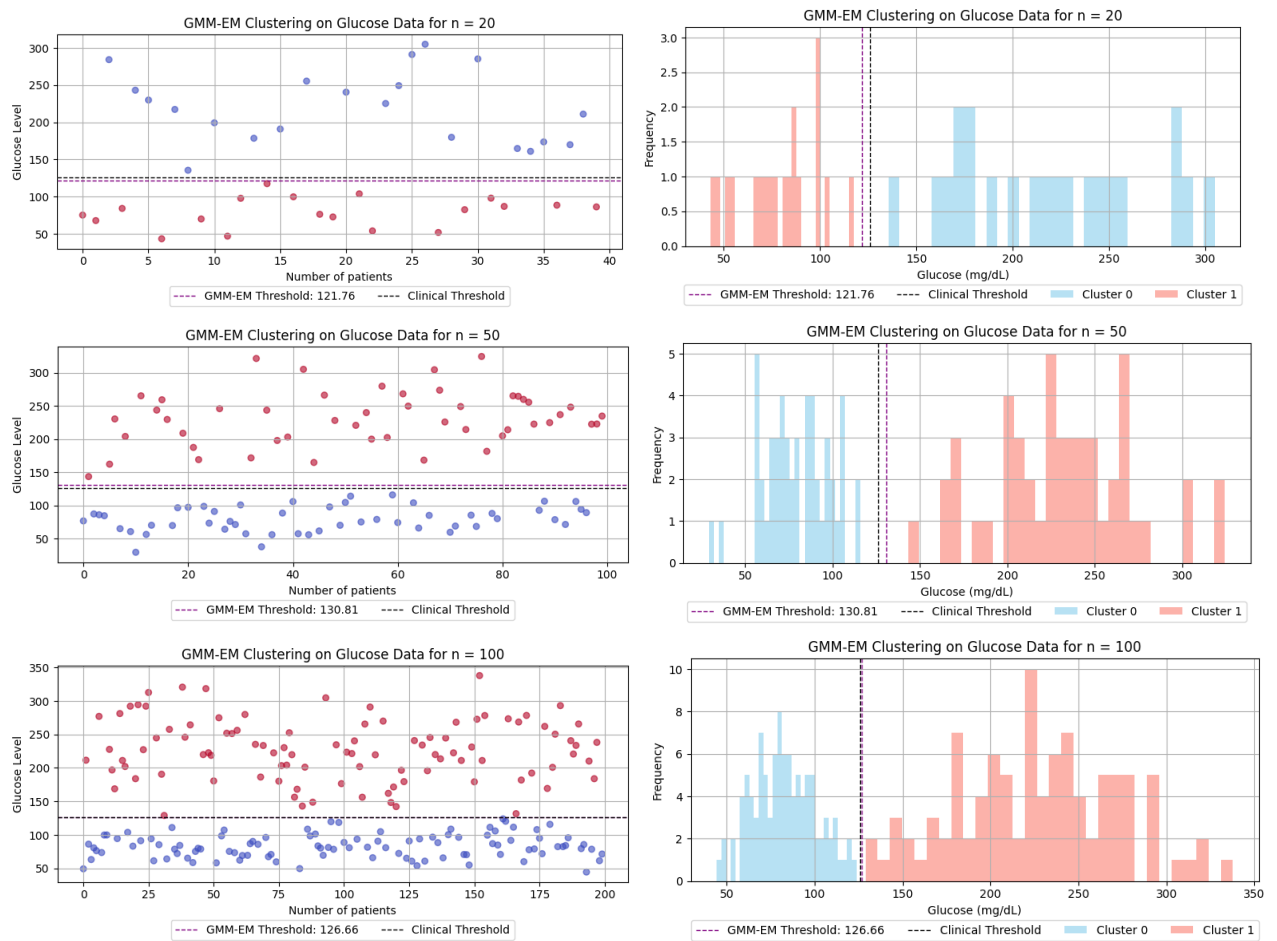


Figure 4 : Distributions and scatter plots displaying the identified clusters and the calculated thresholds after applying GMM-EM clustering for n = 20 (121.76 mg/dL), 50 (130.81 mg/dL) and 100 (126.66 mg/dL). Numbers in bracket are the K-means thresholds for respective n.

Effect of number of data points used:

In case of both K-means and GMM algorithms, for smaller sample sizes like n = 20, the calculated threshold fluctuates significantly and it becomes stable as the sample size increases. This makes the analysis difficult at lower sample sizes. In the case of DBSCAN no matter what sample size we choose we get varied number of clusters.

Conclusion:

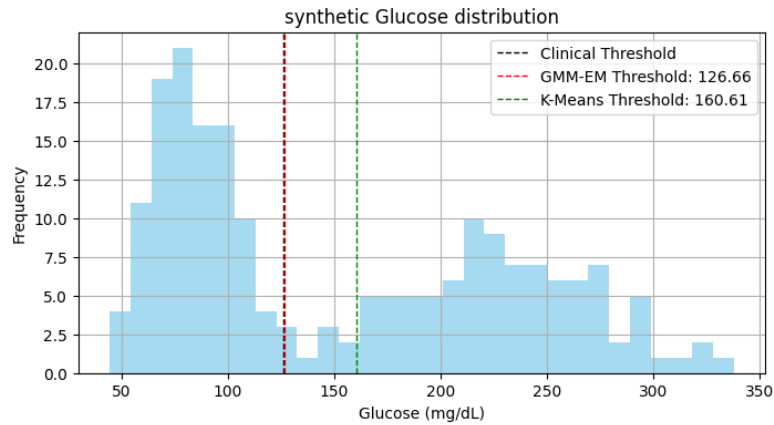


Figure 5: Histogram of synthetic glucose distribution with the thresholds obtained from various clustering algorithms for sample size = 100 each

This experiment evaluates three clustering algorithms namely K-means, DBSCAN and GMM for classifying patients as normal and diabetic from a data which is a mixture of two Gaussians. While K-means is able to classify the data into two clusters but a high threshold as compared to the clinical threshold, DBSCAN is not even able to form 2 clusters and requires a perfect combination of the parameters which are difficult to obtain especially for a 1D dataset like ours. GMM performs best and gives a threshold close enough to the clinically set threshold in this setting.